

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
BOIN12 - No. Pts treated	13.5	6	3.2	1.9	0.9	
BOIN12 - Select %	36.1	14.4	5.9	1.5	0.7	41.4
U-BOIN - No. Pts treated	9.8	5.4	4.3	3.8	3.6	
U-BOIN - Select %	27.8	20.4	16	13.4	9.5	12.9
EffTox - No. Pts treated	8	3.2	3	2	2	
EffTox - Select %	5	4	6	3	6	75
AIDE phase I (Utility 2) - No. Pts treated	16.7	6	1.4	0.3	0.1	
AIDE phase I (Utility 2) - Select %	27.5	16.4	2.6	0.3	0.1	53.1
AIDE phase I (Utility 3) - No. Pts treated	17.5	5.4	0.6	0.1	0	
AIDE phase I (Utility 3) - Select %	24.7	16.1	0.9	0.2	0	58.1
AIDE two-stage (Utility 2) - No. Pts treated	13.2	5.8	3.4	1.3	0.5	
AIDE two-stage (Utility 2) - Select %	34.1	13.1	3.2	0.4	0.3	48.9
AIDE two-stage (Utility 3) - No. Pts treated	13.2	5.7	3.3	1.3	0.5	
AIDE two-stage (Utility 3) - Select %	37.4	11.5	2.1	0.3	0.1	48.6
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
BOIN12 - No. Pts treated	8.6	7.8	8.6	3.6	1.1	
BOIN12 - Select %	29.2	24.7	33.9	9.2	1.5	1.5
U-BOIN - No. Pts treated	8.7	6.9	8.2	4.3	1.8	
U-BOIN - Select %	11.2	11.5	40.6	24.1	11.7	0.9
EffTox - No. Pts treated	22.6	2.5	1.6	1.2	1.7	
EffTox - Select %	72	7	5	6	9	1
AIDE phase I (Utility 2) - No. Pts treated	20.1	7.5	2.1	0.1	0	
AIDE phase I (Utility 2) - Select %	60.4	28.2	10.1	0.5	0	0.8
AIDE phase I (Utility 3) - No. Pts treated	22.1	6.6	1.2	0	0	
AIDE phase I (Utility 3) - Select %	67.9	25.6	5.2	0.1	0	1.2
AIDE two-stage (Utility 2) - No. Pts treated	7.5	6.9	7.7	4.8	2.8	
AIDE two-stage (Utility 2) - Select %	23.6	21.4	38.9	11.9	3.1	1.1
AIDE two-stage (Utility 3) - No. Pts treated	7.6	6.9	7.8	4.8	2.8	
AIDE two-stage (Utility 3) - Select %	27.4	21.6	37.2	10.5	2.2	1.1
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
BOIN12 - No. Pts treated	6.9	11.3	6.4	3.5	1.6	
BOIN12 - Select %	13.2	56	20.6	6.9	1.2	2.1
U-BOIN - No. Pts treated	7.2	10.7	6.7	3.6	1.7	
U-BOIN - Select %	3.8	44.7	24.5	16.2	10	0.8
EffTox - No. Pts treated	18.8	4.8	2.6	1.5	1.9	
EffTox - Select %	49	19	8	7	15	2
AIDE phase I (Utility 2) - No. Pts treated	12.5	15.5	1.5	0.2	0	
AIDE phase I (Utility 2) - Select %	21.8	67.8	6.1	0.8	0	3.5
AIDE phase I (Utility 3) - No. Pts treated	13.9	14.8	0.9	0	0	
AIDE phase I (Utility 3) - Select %	27.2	64.4	3.5	0.1	0	4.8
AIDE two-stage (Utility 2) - No. Pts treated	7	7.3	7.6	4.9	2.8	
AIDE two-stage (Utility 2) - Select %	11	49.8	24	9.1	2.7	3.4
AIDE two-stage (Utility 3) - No. Pts treated	7	7.4	7.6	4.8	2.8	
AIDE two-stage (Utility 3) - Select %	13.2	52.8	21.7	7.3	1.6	3.4

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
BOIN12 - No. Pts treated	5.4	5.9	6.9	8.7	3.1	
BOIN12 - Select %	15.6	14.2	25.6	36.9	7.4	0.3
U-BOIN - No. Pts treated	5.1	6.1	7.4	7.7	3.9	
U-BOIN - Select %	3.7	6	18.4	43.5	28.4	0
EffTox - No. Pts treated	9.4	4.8	5.3	3.7	6.4	
EffTox - Select %	7	14	20	17	40	1
AIDE phase I (Utility 2) - No. Pts treated	8.9	10.4	8.3	2.4	0.1	
AIDE phase I (Utility 2) - Select %	13.2	32.5	39.8	13.9	0.2	0.4
AIDE phase I (Utility 3) - No. Pts treated	11.2	11.1	6.2	1.5	0	
AIDE phase I (Utility 3) - Select %	17.4	42.7	30	9.1	0.1	0.7
AIDE two-stage (Utility 2) - No. Pts treated	3.7	4	6.5	8.1	7.7	
AIDE two-stage (Utility 2) - Select %	5.2	7.1	22.5	44.2	21	0
AIDE two-stage (Utility 3) - No. Pts treated	3.7	3.9	6.6	8.1	7.7	
AIDE two-stage (Utility 3) - Select %	8	7.5	22.7	44.8	17	0
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	-0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
BOIN12 - No. Pts treated	4.5	7.8	6.7	6.7	4.4	
BOIN12 - Select %	7	42.3	25.2	18	7	0.5
U-BOIN - No. Pts treated	5.2	8.5	6.9	5.7	3.9	
U-BOIN - Select %	2.9	37.5	26.5	21.3	11.7	0.1
EffTox - No. Pts treated	12.8	5.9	4.7	2.8	3.5	
EffTox - Select %	22	22	17	13	25	2
AIDE phase I (Utility 2) - No. Pts treated	7.8	17.7	3.6	0.7	0.1	
AIDE phase I (Utility 2) - Select %	7.3	71.2	16.8	3.7	0.6	0.4
AIDE phase I (Utility 3) - No. Pts treated	8.9	18.1	2.5	0.5	0.1	
AIDE phase I (Utility 3) - Select %	7.9	76.4	11.4	2.7	0.2	1.4
AIDE two-stage (Utility 2) - No. Pts treated	4.1	4.5	5.5	6.6	9.3	
AIDE two-stage (Utility 2) - Select %	4.2	36.3	26.2	21.1	12.1	0.1
AIDE two-stage (Utility 3) - No. Pts treated	4.1	4.5	5.5	6.6	9.3	
AIDE two-stage (Utility 3) - Select %	6.3	40.4	26.3	19.2	7.7	0.1
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	-0	-0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
BOIN12 - No. Pts treated	15.9	6.4	3.3	1.8	0.6	
BOIN12 - Select %	64.3	16.7	5.3	1.7	0.4	11.6
U-BOIN - No. Pts treated	11.9	6.1	4.3	3.1	1.9	
U-BOIN - Select %	25.6	17.2	13.9	17.6	14.3	11.4
EffTox - No. Pts treated	25.6	1.1	0.6	0.3	0.3	
EffTox - Select %	85	1	2	1	1	10
AIDE phase I (Utility 2) - No. Pts treated	24.4	2.7	0.3	0.1	0	
AIDE phase I (Utility 2) - Select %	77.8	7.4	0.8	0.1	0	13.9
AIDE phase I (Utility 3) - No. Pts treated	24.6	2.7	0.1	0	0	
AIDE phase I (Utility 3) - Select %	78.3	7.4	0.2	0.1	0	14
AIDE two-stage (Utility 2) - No. Pts treated	15.5	6	3.6	1.6	0.6	
AIDE two-stage (Utility 2) - Select %	61	16.8	5.7	2	0.2	14.3
AIDE two-stage (Utility 3) - No. Pts treated	15.5	6	3.6	1.6	0.5	
AIDE two-stage (Utility 3) - Select %	65.7	13.8	4.5	1.7	0.1	14.2

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
BOIN12 - No. Pts treated	6.9	12.5	5.9	3.1	1.2	
BOIN12 - Select %	12.8	51.1	20.7	9.8	3.4	2.2
U-BOIN - No. Pts treated	7.6	10.7	6.5	3.4	1.4	
U-BOIN - Select %	4.3	48.3	25.8	13.7	6.2	1.7
EffTox - No. Pts treated	22.9	2.8	1.6	1.1	1.4	
EffTox - Select %	72	8	6	4	10	1
AIDE phase I (Utility 2) - No. Pts treated	19.1	10.4	0.4	0	0	
AIDE phase I (Utility 2) - Select %	52.2	45.5	1.5	0	0	0.8
AIDE phase I (Utility 3) - No. Pts treated	19.5	10.1	0.2	0	0	
AIDE phase I (Utility 3) - Select %	54.9	43.1	0.7	0	0	1.3
AIDE two-stage (Utility 2) - No. Pts treated	7.7	6.6	6.7	4.8	4	
AIDE two-stage (Utility 2) - Select %	14.3	43.4	24.6	11.7	4.5	1.5
AIDE two-stage (Utility 3) - No. Pts treated	7.6	6.6	6.7	4.8	4	
AIDE two-stage (Utility 3) - Select %	16.3	47.7	21.5	9.9	3.1	1.5
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
BOIN12 - No. Pts treated	14.5	5.9	3.1	1.7	0.7	
BOIN12 - Select %	46.2	15	5.1	1.6	0.5	31.6
U-BOIN - No. Pts treated	8.9	4.7	4.2	4.3	4.1	
U-BOIN - Select %	14.8	10.2	14.5	17.4	26.7	16.4
EffTox - No. Pts treated	14.3	3.3	2.4	1.5	1.9	
EffTox - Select %	27	11	8	5	9	39
AIDE phase I (Utility 2) - No. Pts treated	18.8	5.3	1	0.2	0	
AIDE phase I (Utility 2) - Select %	46.4	12.8	1.8	0.2	0.1	38.7
AIDE phase I (Utility 3) - No. Pts treated	19.6	4.6	0.5	0.1	0	
AIDE phase I (Utility 3) - Select %	46.9	10.6	1.1	0	0.1	41.3
AIDE two-stage (Utility 2) - No. Pts treated	15.3	5.2	2.9	1	0.3	
AIDE two-stage (Utility 2) - Select %	44.4	13.2	3.4	1.5	0.2	37.3
AIDE two-stage (Utility 3) - No. Pts treated	15.3	5.1	2.9	1	0.4	
AIDE two-stage (Utility 3) - Select %	46.2	11.7	2.8	1.4	0.1	37.8
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	-0	-0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
BOIN12 - No. Pts treated	6.6	5.9	5.7	5.2	6.2	
BOIN12 - Select %	18.9	16.2	22.4	12.6	25.3	4.6
U-BOIN - No. Pts treated	6.8	5.6	6.3	5.2	6	
U-BOIN - Select %	3.8	3.9	13.8	19.1	58	1.4
EffTox - No. Pts treated	9.3	3.9	4.6	3	7.7	
EffTox - Select %	8	8	12	8	56	9
AIDE phase I (Utility 2) - No. Pts treated	11.5	9.3	6.1	2.1	0.4	
AIDE phase I (Utility 2) - Select %	19.1	32.8	27.5	10.6	3	7
AIDE phase I (Utility 3) - No. Pts treated	14.2	9.9	3.9	1	0.2	
AIDE phase I (Utility 3) - Select %	23.2	44.3	18.1	5	1	8.4
AIDE two-stage (Utility 2) - No. Pts treated	6.4	5.4	6.3	4.9	6.2	
AIDE two-stage (Utility 2) - Select %	11.5	14.2	24.5	13.8	29.1	6.9
AIDE two-stage (Utility 3) - No. Pts treated	6.4	5.4	6.4	4.8	6.2	
AIDE two-stage (Utility 3) - Select %	14.6	16	23.7	12.4	26.3	7

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
BOIN12 - No. Pts treated	8.8	10.4	5.1	2.5	1.1	
BOIN12 - Select %	14.6	43.9	16.1	4	2.2	19.2
U-BOIN - No. Pts treated	7.7	9.2	6.5	3.6	2	
U-BOIN - Select %	1.2	32.8	27.1	17	17.3	4.6
EffTox - No. Pts treated	14.5	5.4	3.2	2.2	3.2	
EffTox - Select %	22	19	10	12	29	8
AIDE phase I (Utility 2) - No. Pts treated	12.5	13.8	1.2	0.1	0	
AIDE phase I (Utility 2) - Select %	16.4	53.4	5	0.2	0	25
AIDE phase I (Utility 3) - No. Pts treated	13.3	13.1	0.8	0.1	0	
AIDE phase I (Utility 3) - Select %	16.3	52.5	3.2	0.1	0	27.9
AIDE two-stage (Utility 2) - No. Pts treated	10.2	7.3	5.2	2.8	1.5	
AIDE two-stage (Utility 2) - Select %	11	40.2	15.5	4.9	1.8	26.6
AIDE two-stage (Utility 3) - No. Pts treated	10.2	7.3	5.2	2.8	1.5	
AIDE two-stage (Utility 3) - Select %	11.4	42.7	14	4.2	1.6	26.1
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
BOIN12 - No. Pts treated	14.7	8	3.9	1.7	0.5	
BOIN12 - Select %	64.7	23.1	4.4	0.2	0.1	7.5
U-BOIN - No. Pts treated	11.9	6.6	4.7	3.2	2.8	
U-BOIN - Select %	49.1	19.1	13.5	6.8	8	3.5
EffTox - No. Pts treated	21.8	2.2	1.5	0.8	0.4	
EffTox - Select %	69	5	3	2	1	20
AIDE phase I (Utility 2) - No. Pts treated	22.7	5.6	0.7	0.1	0	
AIDE phase I (Utility 2) - Select %	74.2	18.4	1.6	0	0	5.8
AIDE phase I (Utility 3) - No. Pts treated	24.2	4.3	0.5	0	0	
AIDE phase I (Utility 3) - Select %	79.6	13.4	0.9	0	0	6.1
AIDE two-stage (Utility 2) - No. Pts treated	13.7	8.7	5	1.3	0.2	
AIDE two-stage (Utility 2) - Select %	64.4	24.8	4.3	0.1	0	6.4
AIDE two-stage (Utility 3) - No. Pts treated	13.8	8.6	5	1.3	0.2	
AIDE two-stage (Utility 3) - Select %	68.6	22.1	2.6	0.2	0	6.5
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	-0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
BOIN12 - No. Pts treated	5.4	6.5	5.9	7.9	4.2	
BOIN12 - Select %	20	26.2	13.1	31	9.5	0.2
U-BOIN - No. Pts treated	5.2	6.3	6.4	7.4	4.9	
U-BOIN - Select %	6.1	9.6	6.3	38.7	39.2	0.1
EffTox - No. Pts treated	9.6	4.1	6.1	3.9	5.8	
EffTox - Select %	10	9	24	18	36	4
AIDE phase I (Utility 2) - No. Pts treated	9.3	11.1	5.4	3.9	0.3	
AIDE phase I (Utility 2) - Select %	15.6	40.9	18.3	23.5	1.4	0.3
AIDE phase I (Utility 3) - No. Pts treated	12	11	4.2	2.7	0.1	
AIDE phase I (Utility 3) - Select %	20.8	47.3	15.2	15.5	0.5	0.7
AIDE two-stage (Utility 2) - No. Pts treated	3.9	4	6.4	8.3	7.5	
AIDE two-stage (Utility 2) - Select %	7.9	12.5	13	41.7	24.9	0
AIDE two-stage (Utility 3) - No. Pts treated	3.9	3.9	6.4	8.3	7.5	
AIDE two-stage (Utility 3) - Select %	12.7	13.4	12.1	40.3	21.5	0

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	-0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
BOIN12 - No. Pts treated	9.5	9.1	5.3	3	1.3	
BOIN12 - Select %	18.3	41.2	17.5	5	0.9	17.1
U-BOIN - No. Pts treated	7.9	7.1	6.5	4.5	3	
U-BOIN - Select %	4.3	21.5	27.2	21.4	21.3	4.3
EffTox - No. Pts treated	14.5	5.4	3.2	2	2.8	
EffTox - Select %	27	17	11	11	24	10
AIDE phase I (Utility 2) - No. Pts treated	12.5	13.5	1.8	0.2	0	
AIDE phase I (Utility 2) - Select %	17.3	52.5	7.4	0.6	0.1	22.1
AIDE phase I (Utility 3) - No. Pts treated	13.8	12.7	0.9	0.1	0	
AIDE phase I (Utility 3) - Select %	17.8	53.1	3.5	0.3	0.1	25.2
AIDE two-stage (Utility 2) - No. Pts treated	9.5	7.6	5.6	3.3	1.6	
AIDE two-stage (Utility 2) - Select %	12.2	44.1	15.8	5.2	1.7	21
AIDE two-stage (Utility 3) - No. Pts treated	9.5	7.7	5.6	3.2	1.6	
AIDE two-stage (Utility 3) - Select %	13.7	45.2	14.2	4.8	1.3	20.8
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	-0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
BOIN12 - No. Pts treated	6.6	6.8	5.8	6.6	4.1	
BOIN12 - Select %	28.7	22.2	13.5	24.9	9	1.7
U-BOIN - No. Pts treated	6.5	6.6	5.9	6.3	4.8	
U-BOIN - Select %	11.9	9.5	9.2	33.9	35.1	0.4
EffTox - No. Pts treated	9.4	3.9	5.8	3.6	5.8	
EffTox - Select %	9	9	21	15	36	11
AIDE phase I (Utility 2) - No. Pts treated	11	9.8	5.3	3.4	0.4	
AIDE phase I (Utility 2) - Select %	20.3	37	20.3	18.5	2.4	1.5
AIDE phase I (Utility 3) - No. Pts treated	13.4	11	3.7	1.6	0.2	
AIDE phase I (Utility 3) - Select %	25.7	48.5	14.3	7.8	0.8	2.9
AIDE two-stage (Utility 2) - No. Pts treated	4.9	5.1	7.1	6.6	6.1	
AIDE two-stage (Utility 2) - Select %	12.8	18.3	18.8	30.4	18.1	1.6
AIDE two-stage (Utility 3) - No. Pts treated	4.8	5.2	7.1	6.6	6.1	
AIDE two-stage (Utility 3) - Select %	20.5	18.7	16.8	27.7	14.7	1.6
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	-0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
BOIN12 - No. Pts treated	10.3	8.4	5.5	3.6	2	
BOIN12 - Select %	49.4	30.8	12.2	3.4	1.3	2.9
U-BOIN - No. Pts treated	9	7.6	5.6	4.3	3.5	
U-BOIN - Select %	41.1	23.2	11.8	11.2	12.2	0.5
EffTox - No. Pts treated	14.1	3.2	3.7	2.1	1.8	
EffTox - Select %	26	11	10	9	7	37
AIDE phase I (Utility 2) - No. Pts treated	16.3	8.9	3.2	1.1	0.2	
AIDE phase I (Utility 2) - Select %	48.8	34.9	9.7	3	0.4	3.2
AIDE phase I (Utility 3) - No. Pts treated	18.6	8.8	1.7	0.5	0.1	
AIDE phase I (Utility 3) - Select %	55.1	35.6	3.9	0.9	0	4.5
AIDE two-stage (Utility 2) - No. Pts treated	6.9	7.6	8.2	4.6	2.4	
AIDE two-stage (Utility 2) - Select %	43.5	30.8	16.8	5.3	0.9	2.7
AIDE two-stage (Utility 3) - No. Pts treated	6.9	7.6	8.2	4.6	2.4	
AIDE two-stage (Utility 3) - Select %	53.1	29.4	12.1	2.4	0.3	2.7

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
BOIN12 - No. Pts treated	5	5.7	6.5	7.4	5.4	
BOIN12 - Select %	36.8	26.2	20.4	11.7	4.9	0
U-BOIN - No. Pts treated	6.1	6	6.2	6.4	5.6	
U-BOIN - Select %	30.4	21.6	22.9	17.1	8	0
EffTox - No. Pts treated	20.9	2.3	2.5	1.7	2.1	
EffTox - Select %	66	5	8	7	9	5
AIDE phase I (Utility 2) - No. Pts treated	17.9	7.6	3.2	1	0.3	
AIDE phase I (Utility 2) - Select %	53.1	26.9	13.5	4.8	1.6	0.1
AIDE phase I (Utility 3) - No. Pts treated	19.7	7.4	2.3	0.5	0.1	
AIDE phase I (Utility 3) - Select %	60.6	27.6	8.9	2.6	0.3	0
AIDE two-stage (Utility 2) - No. Pts treated	3.7	3.6	3.8	4.9	14	
AIDE two-stage (Utility 2) - Select %	33.5	22.9	16	16.1	11.5	0
AIDE two-stage (Utility 3) - No. Pts treated	3.7	3.6	3.8	5	13.9	
AIDE two-stage (Utility 3) - Select %	38.5	24.8	16.9	13.8	6	0
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
BOIN12 - No. Pts treated	10	13.6	4.7	1.3	0.2	
BOIN12 - Select %	33.7	55.7	9.2	0.7	0	0.7
U-BOIN - No. Pts treated	10.8	12.1	5.1	1.5	0.4	
U-BOIN - Select %	28.7	50.6	13.6	4.4	2.4	0.3
EffTox - No. Pts treated	26.6	1.8	0.9	0.4	0.1	
EffTox - Select %	91	5	3	1	0	1
AIDE phase I (Utility 2) - No. Pts treated	22.9	6.7	0.3	0	0	
AIDE phase I (Utility 2) - Select %	70.4	28.4	1.1	0	0	0.1
AIDE phase I (Utility 3) - No. Pts treated	23.5	6.3	0.2	0	0	
AIDE phase I (Utility 3) - Select %	73.9	25.4	0.5	0	0	0.2
AIDE two-stage (Utility 2) - No. Pts treated	6.9	10.8	9.1	2.8	0.4	
AIDE two-stage (Utility 2) - Select %	23.5	58	17.6	0.6	0	0.3
AIDE two-stage (Utility 3) - No. Pts treated	6.9	10.7	9.1	2.8	0.4	
AIDE two-stage (Utility 3) - Select %	26.3	59.1	13.8	0.5	0	0.3
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
BOIN12 - No. Pts treated	4.2	5.2	6	6.7	7.9	
BOIN12 - Select %	15.8	19.6	21	15.5	28.1	0
U-BOIN - No. Pts treated	4.2	5.4	6.8	7.1	6.8	
U-BOIN - Select %	7.9	10.8	18.2	15.9	47.2	0
EffTox - No. Pts treated	8.5	3.9	6.2	4.1	6.5	
EffTox - Select %	6	7	21	18	42	6
AIDE phase I (Utility 2) - No. Pts treated	8.2	10	6.8	3.8	1.2	
AIDE phase I (Utility 2) - Select %	14.7	31.3	27.9	18.3	7.6	0.2
AIDE phase I (Utility 3) - No. Pts treated	9.9	11.3	6	2.2	0.7	
AIDE phase I (Utility 3) - Select %	17.8	42.6	25.1	10	4.2	0.3
AIDE two-stage (Utility 2) - No. Pts treated	3.3	3.2	3.8	6.3	13.4	
AIDE two-stage (Utility 2) - Select %	7.8	9.8	9.5	15.4	57.5	0
AIDE two-stage (Utility 3) - No. Pts treated	3.2	3.2	3.9	6.3	13.4	
AIDE two-stage (Utility 3) - Select %	11	12.2	12	13.2	51.6	0

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	-0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
BOIN12 - No. Pts treated	10	12.1	5.4	1.8	0.4	
BOIN12 - Select %	26.7	57	11.2	0.7	0	4.4
U-BOIN - No. Pts treated	7.8	10.3	6.3	3.4	2.1	
U-BOIN - Select %	16.5	39.9	22.5	11.5	9.3	0.3
EffTox - No. Pts treated	17.6	5.9	3.2	1.3	0.6	
EffTox - Select %	45	28	9	6	3	10
AIDE phase I (Utility 2) - No. Pts treated	12.5	15.1	2	0.2	0	
AIDE phase I (Utility 2) - Select %	23.8	64.4	7.9	0.2	0	3.7
AIDE phase I (Utility 3) - No. Pts treated	14	14.5	1.2	0	0	
AIDE phase I (Utility 3) - Select %	26.3	63.7	5	0.2	0	4.8
AIDE two-stage (Utility 2) - No. Pts treated	7.3	11.1	8.5	2.5	0.3	
AIDE two-stage (Utility 2) - Select %	17.2	62.7	15.9	1.1	0	3.1
AIDE two-stage (Utility 3) - No. Pts treated	7.3	11.1	8.5	2.5	0.3	
AIDE two-stage (Utility 3) - Select %	19.9	61.8	14.2	1.1	0	3
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	-0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
BOIN12 - No. Pts treated	6.7	10.7	6.8	4.1	1.7	
BOIN12 - Select %	16.7	58	18.5	5.2	0.7	0.9
U-BOIN - No. Pts treated	6.1	10.8	6.8	4	2.3	
U-BOIN - Select %	10	45	16.8	14	14	0.2
EffTox - No. Pts treated	13.5	6.6	5.1	2	1.7	
EffTox - Select %	24	30	20	7	10	9
AIDE phase I (Utility 2) - No. Pts treated	8.1	17	4.1	0.7	0.1	
AIDE phase I (Utility 2) - Select %	9.6	67.9	18.9	2.9	0.2	0.5
AIDE phase I (Utility 3) - No. Pts treated	9.7	17	2.9	0.4	0	
AIDE phase I (Utility 3) - Select %	11.9	72.2	13.4	1.8	0	0.7
AIDE two-stage (Utility 2) - No. Pts treated	4.3	6.9	10	6	2.7	
AIDE two-stage (Utility 2) - Select %	9.6	49.5	30	9.1	1.6	0.2
AIDE two-stage (Utility 3) - No. Pts treated	4.3	6.8	10.1	6	2.8	
AIDE two-stage (Utility 3) - Select %	12.8	51.1	26.7	8.3	0.9	0.2
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
BOIN12 - No. Pts treated	19.9	2.6	0.4	0	0	
BOIN12 - Select %	58.5	0.9	0.1	0	0	40.5
U-BOIN - No. Pts treated	17	4	1.6	1	0.8	
U-BOIN - Select %	49.7	8.4	7.2	4.9	5.8	24
EffTox - No. Pts treated	25	0.3	0.1	0	0	
EffTox - Select %	72	0	0	0	0	28
AIDE phase I (Utility 2) - No. Pts treated	23	0.5	0	0	0	
AIDE phase I (Utility 2) - Select %	62.1	0.4	0	0	0	37.5
AIDE phase I (Utility 3) - No. Pts treated	23.1	0.4	0	0	0	
AIDE phase I (Utility 3) - Select %	61.9	0.2	0	0	0	37.9
AIDE two-stage (Utility 2) - No. Pts treated	19.7	2.9	0.6	0	0	
AIDE two-stage (Utility 2) - Select %	58	1.6	0	0	0	40.4
AIDE two-stage (Utility 3) - No. Pts treated	19.7	2.9	0.6	0	0	
AIDE two-stage (Utility 3) - Select %	58.2	1.5	0	0	0	40.3

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	-0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
BOIN12 - No. Pts treated	11.6	11.4	4.4	2	0.3	
BOIN12 - Select %	40.8	50.2	3.8	0.6	0	4.6
U-BOIN - No. Pts treated	8.3	9.7	5	4.6	2.4	
U-BOIN - Select %	25.2	30.3	10	23.2	11.1	0.2
EffTox - No. Pts treated	17	5.9	3.3	1.2	0.5	
EffTox - Select %	39	29	10	4	2	16
AIDE phase I (Utility 2) - No. Pts treated	13	14.7	2	0.3	0	
AIDE phase I (Utility 2) - Select %	27.4	62.5	8.2	0.6	0	1.3
AIDE phase I (Utility 3) - No. Pts treated	14.3	14.2	1.4	0.1	0	
AIDE phase I (Utility 3) - Select %	28.9	64.2	4.6	0.2	0	2.1
AIDE two-stage (Utility 2) - No. Pts treated	7.2	13.1	7.7	1.7	0.2	
AIDE two-stage (Utility 2) - Select %	23.1	64.4	10.4	0.9	0	1.2
AIDE two-stage (Utility 3) - No. Pts treated	7.3	13.1	7.7	1.7	0.2	
AIDE two-stage (Utility 3) - Select %	28.4	62.3	7.2	1	0	1.1
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
BOIN12 - No. Pts treated	16.1	4.7	1.1	0.1	0	
BOIN12 - Select %	24.5	3	0	0	0	72.5
U-BOIN - No. Pts treated	13.3	4.7	2.9	2.6	2.6	
U-BOIN - Select %	47.6	18.1	5.9	1.9	0.5	26
EffTox - No. Pts treated	8	3.1	1.7	0.3	0	
EffTox - Select %	5	4	4	1	0	86
AIDE phase I (Utility 2) - No. Pts treated	17.3	4	0.3	0	0	
AIDE phase I (Utility 2) - Select %	20.9	5	0.1	0	0	74
AIDE phase I (Utility 3) - No. Pts treated	17.6	3.5	0.2	0	0	
AIDE phase I (Utility 3) - Select %	19.5	5	0.1	0	0	75.4
AIDE two-stage (Utility 2) - No. Pts treated	16.7	4.2	0.6	0	0	
AIDE two-stage (Utility 2) - Select %	22.6	3.2	0	0	0	74.2
AIDE two-stage (Utility 3) - No. Pts treated	16.7	4.2	0.7	0	0	
AIDE two-stage (Utility 3) - Select %	22.8	2.8	0	0	0	74.4
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
BOIN12 - No. Pts treated	3.2	3.4	4.5	12.5	6.3	
BOIN12 - Select %	2.6	2.1	6.9	62.8	25.6	0
U-BOIN - No. Pts treated	3.4	3.9	5	9.7	8.3	
U-BOIN - Select %	1.2	1.7	7.3	60.4	29.4	0
EffTox - No. Pts treated	8.1	3.3	4.6	4	9.9	
EffTox - Select %	5	5	10	18	63	1
AIDE phase I (Utility 2) - No. Pts treated	7.6	8.3	8.6	5.2	0.3	
AIDE phase I (Utility 2) - Select %	10.8	20.4	36.9	29.5	2.2	0.2
AIDE phase I (Utility 3) - No. Pts treated	8.9	9.3	7.7	3.7	0.3	
AIDE phase I (Utility 3) - Select %	12.1	28.5	34.8	22.3	1.8	0.5
AIDE two-stage (Utility 2) - No. Pts treated	3.1	3	3.2	4.6	16	
AIDE two-stage (Utility 2) - Select %	1.2	1.5	4.3	38.7	54.3	0
AIDE two-stage (Utility 3) - No. Pts treated	3.1	3	3.2	4.6	16	
AIDE two-stage (Utility 3) - Select %	2.1	2.5	6.1	43	46.3	0

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
BOIN12 - No. Pts treated	10.2	9.8	7.5	2.2	0.3	
BOIN12 - Select %	43.2	36	20.1	0.6	0	0.1
U-BOIN - No. Pts treated	9.7	9	7.2	2.9	1.2	
U-BOIN - Select %	30	25.3	26.2	9.9	8.5	0.1
EffTox - No. Pts treated	22.9	3.3	2	1	0.3	
EffTox - Select %	77	12	5	3	1	3
AIDE phase I (Utility 2) - No. Pts treated	19.4	8.4	2.1	0.1	0	
AIDE phase I (Utility 2) - Select %	57.1	33.2	9.4	0.1	0	0.2
AIDE phase I (Utility 3) - No. Pts treated	20.7	7.9	1.4	0	0	
AIDE phase I (Utility 3) - Select %	63.3	30.2	6.2	0.1	0	0.2
AIDE two-stage (Utility 2) - No. Pts treated	5.8	9.5	10.9	3.3	0.4	
AIDE two-stage (Utility 2) - Select %	30.9	34.2	32.7	1.9	0	0.3
AIDE two-stage (Utility 3) - No. Pts treated	5.8	9.5	11	3.3	0.4	
AIDE two-stage (Utility 3) - Select %	36	33	29	1.7	0	0.3
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	-0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
BOIN12 - No. Pts treated	9.4	8.9	8.1	2.9	0.4	
BOIN12 - Select %	31.1	32.3	32.3	0.9	0	3.4
U-BOIN - No. Pts treated	7.9	7.6	8	4.1	2.1	
U-BOIN - Select %	15.8	22.3	36.7	12.9	11.2	1.1
EffTox - No. Pts treated	17	4.9	3.8	2.1	1	
EffTox - Select %	38	25	18	8	4	8
AIDE phase I (Utility 2) - No. Pts treated	15	11.4	3.1	0.2	0	
AIDE phase I (Utility 2) - Select %	35	45	15.4	0.5	0	4.1
AIDE phase I (Utility 3) - No. Pts treated	16.8	10.4	2.3	0.1	0	
AIDE phase I (Utility 3) - Select %	41.5	42.1	10.8	0.2	0	5.4
AIDE two-stage (Utility 2) - No. Pts treated	7.7	8.3	9.6	3.5	0.4	
AIDE two-stage (Utility 2) - Select %	23.5	36.5	32.3	4	0	3.7
AIDE two-stage (Utility 3) - No. Pts treated	7.6	8.3	9.7	3.5	0.5	
AIDE two-stage (Utility 3) - Select %	26.7	37.2	28.3	4.1	0.1	3.6
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	-0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
BOIN12 - No. Pts treated	7.5	9	9.1	3.3	0.8	
BOIN12 - Select %	22.1	32.1	40.3	3.5	0.2	1.8
U-BOIN - No. Pts treated	6.9	7.9	8.9	4.4	1.8	
U-BOIN - Select %	9.9	20.2	39.6	15.7	14.4	0.2
EffTox - No. Pts treated	16.6	5.1	3.7	2.2	1.7	
EffTox - Select %	36	27	15	12	6	4
AIDE phase I (Utility 2) - No. Pts treated	13.6	11.8	4.2	0.2	0	
AIDE phase I (Utility 2) - Select %	31.4	45.1	21.7	0.5	0	1.3
AIDE phase I (Utility 3) - No. Pts treated	15.2	11.7	2.8	0.1	0	
AIDE phase I (Utility 3) - Select %	35.5	46.8	14.4	0.4	0.1	2.8
AIDE two-stage (Utility 2) - No. Pts treated	5.6	7.3	10.7	5.1	1.3	
AIDE two-stage (Utility 2) - Select %	16	29	46.7	7	0.6	0.7
AIDE two-stage (Utility 3) - No. Pts treated	5.6	7.3	10.6	5.1	1.2	
AIDE two-stage (Utility 3) - Select %	19.4	30.6	43.5	5.4	0.3	0.8

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
BOIN12 - No. Pts treated	11.5	11.1	4	1.3	0.2	
BOIN12 - Select %	25.1	50.6	4.4	0.5	0	19.4
U-BOIN - No. Pts treated	8.4	9	5.5	3.5	2.5	
U-BOIN - Select %	16.7	39.7	15.4	12.7	11.3	4.2
EffTox - No. Pts treated	15.7	6.1	3.3	1.4	0.7	
EffTox - Select %	36	29	8	5	4	19
AIDE phase I (Utility 2) - No. Pts treated	12.7	14.3	1	0.1	0	
AIDE phase I (Utility 2) - Select %	18.3	57	2.6	0	0	22.1
AIDE phase I (Utility 3) - No. Pts treated	13.8	13.4	0.7	0	0	
AIDE phase I (Utility 3) - Select %	19.6	56.5	1	0	0	22.9
AIDE two-stage (Utility 2) - No. Pts treated	10.6	10.7	5.2	1.1	0.1	
AIDE two-stage (Utility 2) - Select %	21.9	50.5	6.1	0.4	0	21.1
AIDE two-stage (Utility 3) - No. Pts treated	10.6	10.6	5.3	1.1	0.1	
AIDE two-stage (Utility 3) - Select %	22.6	51.2	4.8	0.4	0	21
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
BOIN12 - No. Pts treated	17.1	8.8	2.8	0.6	0	
BOIN12 - Select %	71	24.8	1.2	0	0	3
U-BOIN - No. Pts treated	16.2	8.8	3	1	0.6	
U-BOIN - Select %	61.5	24.3	4.9	3.8	3.6	1.9
EffTox - No. Pts treated	28.3	0.7	0.3	0.1	0	
EffTox - Select %	96	1	0	0	0	3
AIDE phase I (Utility 2) - No. Pts treated	26.7	2.6	0.1	0	0	
AIDE phase I (Utility 2) - Select %	87.8	9.6	0	0	0	2.6
AIDE phase I (Utility 3) - No. Pts treated	27.1	2.3	0.1	0	0	
AIDE phase I (Utility 3) - Select %	89.8	7.7	0	0	0	2.5
AIDE two-stage (Utility 2) - No. Pts treated	13	10.8	4.9	0.6	0	
AIDE two-stage (Utility 2) - Select %	66	28.9	2.4	0	0	2.7
AIDE two-stage (Utility 3) - No. Pts treated	13	10.8	4.9	0.6	0	
AIDE two-stage (Utility 3) - Select %	68.4	27.2	1.7	0	0	2.7
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	-0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
BOIN12 - No. Pts treated	5.2	6.1	6.7	8.8	3.2	
BOIN12 - Select %	22.4	21.9	19.1	33.7	2.9	0
U-BOIN - No. Pts treated	5.4	6.6	7.3	7.4	3.7	
U-BOIN - Select %	12.4	14.7	17.5	38.8	16.6	0
EffTox - No. Pts treated	15.9	3.5	4.6	2.8	2.8	
EffTox - Select %	39	14	20	14	11	3
AIDE phase I (Utility 2) - No. Pts treated	13.2	9.7	5	2	0.1	
AIDE phase I (Utility 2) - Select %	34.9	31.7	20.4	12	0.3	0.7
AIDE phase I (Utility 3) - No. Pts treated	15.5	9.8	3.5	1.2	0	
AIDE phase I (Utility 3) - Select %	42.7	35.9	14.7	6	0.3	0.4
AIDE two-stage (Utility 2) - No. Pts treated	3.7	3.7	6.1	9.7	6.8	
AIDE two-stage (Utility 2) - Select %	14.4	16	14.6	42.2	12.8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.7	3.7	6.1	9.7	6.8	
AIDE two-stage (Utility 3) - Select %	19.5	17.1	14.9	38	10.5	0

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
BOIN12 - No. Pts treated	3.8	12.3	8.1	4.6	1.1	
BOIN12 - Select %	3.1	41.6	34.4	20.1	0.8	0
U-BOIN - No. Pts treated	4.5	7.6	7.8	7.5	3	
U-BOIN - Select %	4.6	39.7	34.4	20	1.3	0
EffTox - No. Pts treated	20.7	2.9	3.1	2.3	1	
EffTox - Select %	62	9	15	12	3	0
AIDE phase I (Utility 2) - No. Pts treated	16.7	12.5	0.7	0.1	0	
AIDE phase I (Utility 2) - Select %	44.4	52.2	2.9	0.5	0	0
AIDE phase I (Utility 3) - No. Pts treated	17.2	12.3	0.5	0	0	
AIDE phase I (Utility 3) - Select %	46.9	50.7	2.1	0.2	0	0.1
AIDE two-stage (Utility 2) - No. Pts treated	3.5	3.4	4.6	10.1	8.5	
AIDE two-stage (Utility 2) - Select %	6.9	32.5	24.6	29.2	6.8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.5	3.4	4.6	10	8.6	
AIDE two-stage (Utility 3) - Select %	9.5	38.6	23.8	23.6	4.5	0
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
BOIN12 - No. Pts treated	4.3	5.9	9.5	6.7	3.6	
BOIN12 - Select %	14.5	20.5	46.3	15	3.7	0
U-BOIN - No. Pts treated	4.4	6	9.8	6.6	3.4	
U-BOIN - Select %	6.5	16.9	45.1	17.9	13.6	0
EffTox - No. Pts treated	12.1	4.7	5.9	3.6	3.2	
EffTox - Select %	19	18	30	14	15	4
AIDE phase I (Utility 2) - No. Pts treated	9.9	10.5	7.6	1.7	0.2	
AIDE phase I (Utility 2) - Select %	18.7	34	36.1	9.3	1.4	0.5
AIDE phase I (Utility 3) - No. Pts treated	11.5	11.3	5.8	1.2	0.1	
AIDE phase I (Utility 3) - Select %	23.8	41.5	28	5.8	0.6	0.3
AIDE two-stage (Utility 2) - No. Pts treated	3.3	3.3	5.3	7.8	10.3	
AIDE two-stage (Utility 2) - Select %	10.9	16.9	32.9	25.5	13.8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.3	3.4	5.2	7.8	10.3	
AIDE two-stage (Utility 3) - Select %	15	19.1	33.3	23.7	8.9	0
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
BOIN12 - No. Pts treated	4.5	5.6	7.3	8.2	4.4	
BOIN12 - Select %	14.7	15.1	28.2	30.3	11.7	0
U-BOIN - No. Pts treated	3.9	5	8.7	8	4.5	
U-BOIN - Select %	6	7.9	27.8	28.5	29.8	0
EffTox - No. Pts treated	8.9	4.1	7.1	4.5	5.3	
EffTox - Select %	6	9	34	21	29	1
AIDE phase I (Utility 2) - No. Pts treated	7.8	8.5	9.1	4.3	0.3	
AIDE phase I (Utility 2) - Select %	11.7	22.1	40.2	24.1	1.6	0.3
AIDE phase I (Utility 3) - No. Pts treated	9.8	9.2	7.5	3.4	0.2	
AIDE phase I (Utility 3) - Select %	15.8	28.6	34.9	19.4	0.8	0.5
AIDE two-stage (Utility 2) - No. Pts treated	3.2	3.1	4.7	8	11	
AIDE two-stage (Utility 2) - Select %	6.5	6.7	13.9	41.1	31.8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.2	3.1	4.6	8	11.1	
AIDE two-stage (Utility 3) - Select %	9.6	7.5	15.5	40.7	26.7	0

Phase I/II Operating Characteristics

Method comparison - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	-0	0	-0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
BOIN12 - No. Pts treated	7.4	8	7.6	4.7	2.3	
BOIN12 - Select %	28.2	22.3	35.3	10.8	2.8	0.6
U-BOIN - No. Pts treated	5.8	7.4	8.1	5.3	3.5	
U-BOIN - Select %	10.7	10.4	28.2	23	27.7	0
EffTox - No. Pts treated	10.1	5.9	6.4	2.9	3.4	
EffTox - Select %	12	23	27	11	16	10
AIDE phase I (Utility 2) - No. Pts treated	9.2	10.4	8.9	1.3	0.1	
AIDE phase I (Utility 2) - Select %	14.9	32.1	45.6	6.4	0.6	0.4
AIDE phase I (Utility 3) - No. Pts treated	11.9	10.9	6.4	0.8	0.1	
AIDE phase I (Utility 3) - Select %	20.8	41	33.7	3.7	0.1	0.7
AIDE two-stage (Utility 2) - No. Pts treated	4	5.8	9.6	6.3	4.3	
AIDE two-stage (Utility 2) - Select %	11.3	18	49.6	15.6	5.5	0
AIDE two-stage (Utility 3) - No. Pts treated	4	5.8	9.6	6.3	4.3	
AIDE two-stage (Utility 3) - Select %	14.7	19.3	47.7	14.4	3.9	0
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
BOIN12 - No. Pts treated	5.3	13.8	6.8	2.9	1.1	
BOIN12 - Select %	10.5	62.5	21.7	4.1	1.2	0
U-BOIN - No. Pts treated	5.5	12.8	7.8	3	1	
U-BOIN - Select %	9.9	61.2	20.2	5.9	2.8	0
EffTox - No. Pts treated	21.4	4.3	2.7	1.1	0.4	
EffTox - Select %	66	21	8	4	2	0
AIDE phase I (Utility 2) - No. Pts treated	16.7	12.3	1	0.1	0	
AIDE phase I (Utility 2) - Select %	45.2	50.3	4.2	0.2	0	0.1
AIDE phase I (Utility 3) - No. Pts treated	17.6	11.6	0.7	0.1	0	
AIDE phase I (Utility 3) - Select %	48.9	47.9	2.7	0.4	0	0.1
AIDE two-stage (Utility 2) - No. Pts treated	3.6	5.6	9.9	6.7	4.2	
AIDE two-stage (Utility 2) - Select %	12.5	39.1	34.2	11.4	2.8	0
AIDE two-stage (Utility 3) - No. Pts treated	3.6	5.6	9.9	6.7	4.1	
AIDE two-stage (Utility 3) - Select %	16.1	42.6	29.6	9.8	1.9	0
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
BOIN12 - No. Pts treated	9.7	9.1	5.1	3.7	1.8	
BOIN12 - Select %	29.9	38.9	13.5	6.3	2.8	8.6
U-BOIN - No. Pts treated	8	7.3	5.6	5	4.1	
U-BOIN - Select %	9.3	17.6	18.3	22.9	31.4	0.5
EffTox - No. Pts treated	13.3	5.3	4	2	2.5	
EffTox - Select %	22	20	15	7	18	19
AIDE phase I (Utility 2) - No. Pts treated	12.1	13.7	2.8	0.6	0.1	
AIDE phase I (Utility 2) - Select %	18.3	58.6	11.9	2	0.4	8.8
AIDE phase I (Utility 3) - No. Pts treated	13.8	13.4	1.5	0.4	0	
AIDE phase I (Utility 3) - Select %	21.2	60.8	6	1.5	0	10.5
AIDE two-stage (Utility 2) - No. Pts treated	7.9	9.5	6.6	3.5	1.8	
AIDE two-stage (Utility 2) - Select %	17.1	50.6	15.2	7.3	2.5	7.3
AIDE two-stage (Utility 3) - No. Pts treated	7.9	9.4	6.7	3.5	1.8	
AIDE two-stage (Utility 3) - Select %	20.4	51.1	12.5	6.8	1.5	7.7
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	-0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
BOIN12 - No. Pts treated	11.2	10.1	4.5	2.4	1.1	
BOIN12 - Select %	39.2	39	8.9	1.5	0.5	10.9
U-BOIN - No. Pts treated	8.7	8.4	5.3	4.3	3.4	
U-BOIN - Select %	25.3	25.9	13.7	19.8	15.2	0.1
EffTox - No. Pts treated	13.6	6	4.2	1.5	1.2	
EffTox - Select %	22	26	15	6	7	25
AIDE phase I (Utility 2) - No. Pts treated	11.8	14.8	2.7	0.4	0.1	
AIDE phase I (Utility 2) - Select %	20.4	64.6	11.3	0.7	0	3
AIDE phase I (Utility 3) - No. Pts treated	13.2	14.7	1.6	0.2	0	
AIDE phase I (Utility 3) - Select %	20.9	68.5	5.9	0.5	0	4.2
AIDE two-stage (Utility 2) - No. Pts treated	7.3	11.4	7.2	2.8	1.1	
AIDE two-stage (Utility 2) - Select %	20.1	59.1	14.3	3.3	0.1	3.1
AIDE two-stage (Utility 3) - No. Pts treated	7.3	11.4	7.2	2.8	1.1	
AIDE two-stage (Utility 3) - Select %	23.8	58.6	12.1	2.6	0.1	2.8